

PAPER REBCO Conductor and Stabilizer Materials Under Neutron Irradiation: The Un-Doped-Tape Rule, Graphene-Reinforced Stabilizers and V_2O_3 Self-Protection for a Spherical-Tokamak Breeder

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Abstract

The high-field REBCO magnet of a compact spherical-tokamak breeder is usually de-risked on field and stress; its survival in a *neutron* environment is a separate materials question, and the answer changes the conductor specification. The Kronos Hyperion centrepost sees a peak field of 16.84 T and a fast-neutron fluence budget capped at a frozen $3.9 \times 10^{22} \text{ n m}^{-2}$, while its physically distinct D-³He tandem-mirror burner plug operates at 26.49 T; both are wound from rare-earth barium copper oxide (REBCO) coated conductor. We formalise and quantify four coupled conductor-materials results, each a frozen anchor in the Kronos physics de-risking register. First, a fast-neutron-lifetime *selection rule*: for fusion fluence the plain, un-doped tape outlasts the artificial-pinning-centre (APC) tape, whose engineered defects are optimised for in-field critical current, because irradiation supplies its own pinning and the doped tape reaches its critical-current break-even knee sooner — an inversion of the room-temperature selection. A two-population pinning model with a displacement-damage sink reproduces break-even fluences of $\Phi_{\text{un}}^* = 5.64 \times 10^{22} \text{ n m}^{-2}$ against $\Phi_{\text{APC}}^* = 3.82 \times 10^{22} \text{ n m}^{-2}$ ($\times 1.48$), and a 4 000-sample Monte-Carlo confirms the ordering holds with probability 1.00. Displacement-cascade molecular dynamics and DFT+ U locate the limiter on the oxygen sublattice, which absorbs $\approx 78\%$ of the displacements. Second, a J_c -onset-knee screening surrogate that ranks candidate tapes by the break-even fluence without a full reactor campaign. Third, a graphene-reinforced ultra-pure aluminium stabiliser whose saturating high-field magnetoresistance retains a low-loss current bypass to 30–35 T where copper's non-saturating magnetoresistance does not. Fourth, a vanadium-sesquioxide (V_2O_3) self-protecting interlayer whose Mott metal-insulator transition acts as a temperature-triggered turn-to-turn resistance — insulating at the ~ 20 K operating point, conducting on a hot spot — escaping the no-insulation charge-time-versus-protection trade. We close the quench budget: an adiabatic hot-spot integral gives a 3.67 s margin, and a fused strain-rate/magnetisation precursor fires $34\times$ earlier than terminal voltage, holding the peak temperature to 63 K. Together these define a conductor architecture for a neutron environment, distinct from the magnet field/stress design.

Keywords: REBCO coated conductor; fast-neutron irradiation; critical-current degradation; artificial pinning centres; graphene-aluminium stabiliser; V_2O_3 Mott interlayer; quench protection; spherical-tokamak breeder

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1 Introduction

The economic and physics case for the compact, high-field spherical tokamak rests on a high on-axis and, more demanding, a high on-conductor field [1, 2, 3]. In the Kronos Hyperion breeder the toroidal-field return is a single-turn, demountable, water- or helium-cooled centrepost carrying the plasma current $I_p = 9.66$ MA at major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, elongation $\kappa = 2.0$ and negative triangularity $\delta = -0.30$, delivering $Q = 3.076$ and $P_{\text{fus}} = 85.04$ MW with an on-axis field $B_0 = 8$ T and a peak centrepost field of 16.84 T. The physically distinct D-³He tandem-mirror burner uses a separate high-field plug coil at 26.49 T. Both magnets are wound from REBCO (REBa₂Cu₃O_{7- δ}) coated conductor, the only conductor class that reaches these fields at a tractable temperature [4, 5, 7].

A fusion magnet’s conductor is chosen for field and stress. But at the centrepost it must also survive years of fast-neutron fluence at cryogenic temperature, an environment no accelerator or NMR magnet imposes. Fast neutrons displace atoms in the REBCO layer, and the resulting defects both *help* — adding flux-pinning sites that raise the in-field critical current at low dose — and *harm* — eventually degrading the superconducting connectivity and suppressing the critical temperature [8, 9, 10]. The competition between these two effects, and the differing starting points of a plain tape and an engineered artificial-pinning-centre (APC) tape, sets which conductor lives longer under fusion fluence. This paper reports the conductor-materials results behind the Kronos magnet, and shows that the neutron environment selects a different tape, stabiliser and quench-protection scheme than a non-nuclear optimisation would.

1.1 Prior art

Neutron-irradiation studies of REBCO coated conductors — reactor and spallation campaigns on commercial tapes [8, 9], and ion- and light-particle surrogates [10] — establish that J_c first rises then falls with fluence, with a peak near 10^{22} – 10^{23} n m⁻² depending on tape, temperature and field. Selvamanickam and co-workers and others engineered APC tapes (BaZrO₃, BaHfO₃ nanocolumns) that raise the unirradiated in-field J_c [7], and this is the correct choice for a non-nuclear high-field magnet. The high-field magnet-technology programmes — SPARC/ARC, the 45.5 T LBNL/NHMFL insert, VIPER-cable fusion conductors — have converged on no-insulation (NI) or partial-insulation (PI) winding for self-protection [6, 5, 3], but NI trades a valuable protection margin against an impractically long magnetisation charge time, and quench detection in HTS is hard because the normal-zone propagation velocity (NZPV) is two orders of magnitude below that of low-temperature superconductors [21, 22, 23]. The microscopic mechanism of REBCO radiation tolerance — which sublattice is displaced, and how chain-oxygen loss degrades J_c — is a first-principles materials question that the companion paper treats by DFT+ U and displacement-cascade molecular dynamics [2.18].

1.2 Contribution

Against this background we contribute a conductor-materials specification for a neutron environment, stated as governing equations and quantified against the frozen anchors of the de-risking register^[R0]:

1. A **fast-neutron-lifetime selection rule** (§2.1, Figs. 1–2). A two-population strong-pinning model with an irradiation-driven defect source and a displacement-damage sink predicts that the un-doped tape reaches its critical-current break-even fluence *later* than the APC tape, $\Phi_{\text{un}}^* = 5.64 \times 10^{22}$ n m⁻² versus $\Phi_{\text{APC}}^* = 3.82 \times 10^{22}$ n m⁻² ($\times 1.48$), and a 4 000-sample Monte-Carlo shows the ordering is robust with probability 1.00.
2. A **displacement-damage partition** (§2.2, Fig. 3) built on the Norgett–Robinson–Torrens (NRT) model and cascade molecular dynamics, placing $\approx 78\%$ of the atomic displacements on the oxygen sublattice and identifying chain-oxygen loss as the J_c limiter.
3. A **J_c -onset-knee screening surrogate** (§2.3) that ranks candidate tapes by the break-even fluence, providing a screening metric without a full reactor-irradiation campaign, together with an honest statement of the in-field operating-point tension (Fig. 4).
4. A **stabiliser and self-protecting interlayer** (§2.4–§2.5, Figs. 5–9): a graphene-reinforced ultra-pure aluminium bypass whose saturating magnetoresistance beats copper at 30–35 T, and a V₂O₃ interlayer whose Mott transition provides passive, temperature-triggered turn-to-turn shunting that escapes the NI/PI trade.

All computation was performed on the NVIDIA GPU HPC campaign (§3). We separate what is reactor-qualified from what is a screening result or a material-property target throughout, and

state a single honest gate in §6. The companion set fixes the surrounding context: the consumable-centrepost neutronics and coil integrity [2.16], the fast-fluence/dpa life-limiter and shielding [2.26], the first-principles radiation tolerance [2.18], the fused NV-diamond quench detection [2.19], and the 26.49 T burner plug coil [2.21]; the REBCO field envelope for both magnets is set in [1.3].

2 Governing equations and the formal problem

We assemble the conductor-materials model as four coupled sub-problems: critical current under fluence (§2.1), the microscopic displacement source (§2.2), the stabiliser conductivity (§2.4), and the self-protecting interlayer and quench budget (§2.5–§2.6). Throughout, Φ denotes the accumulated fast-neutron fluence ($E_n > 0.1$ MeV) in n m^{-2} and T the local temperature.

2.1 Critical current under fast-neutron fluence and the lifetime rule

The usable life of a REBCO tape ends when neutron-induced damage has degraded its critical current below the design margin at the peak-field point. In the damage-dominated regime the critical-current density falls with fluence approximately exponentially,

$$J_c(\Phi) = J_c^0 \exp\left(-\frac{\Phi}{\Phi_0}\right), \quad (1)$$

where Φ_0 is the characteristic degradation fluence. The lifetime rule is the fluence at which J_c falls to the operating current density J_{op} at the peak field,

$$\Phi_{\text{life}} = \Phi_0 \ln \frac{J_c^0}{J_{\text{op}}} = \Phi_0 \ln(m_0), \quad m_0 \equiv J_c^0/J_{\text{op}}, \quad (2)$$

so raising the design margin m_0 buys life only logarithmically. This is why the neutron shield, not the conductor margin, is the primary lifetime lever [2.26], and why raising Φ_0 — the materials lever — is pursued in parallel.

Equation (1) omits the low-dose *rise* of J_c . To capture the full non-monotone behaviour that governs the selection between tapes we resolve the critical current from the pinning-force balance. In the critical state the Lorentz force density on the flux lattice equals the volume pinning force,

$$J_c(B, \Phi) B = F_p(B, \Phi), \quad (3)$$

and, in a strong, sparse-pin (direct-summation) limit, the volume pinning force is the areal density of active pins times the elementary pinning force,

$$F_p(B, \Phi) = n_p(\Phi) f_p(B) \zeta, \quad (4)$$

with ζ a dimensionless collective-pinning efficiency. Fast neutrons create cascade defects (nanoscale amorphous zones and point-defect clusters) that act as additional pins; the areal density evolves toward a saturation n_{sat} under a first-order accumulation law,

$$\frac{dn_p}{d\Phi} = \sigma_d \left(1 - \frac{n_p}{n_{\text{sat}}}\right) \implies n_p(\Phi) = n_p^0 + (n_{\text{sat}} - n_p^0) \left[1 - e^{-\Phi/\Phi_r}\right], \quad (5)$$

where $\Phi_r = 1/(\sigma_d/n_{\text{sat}})$ is the pinning-saturation fluence and n_p^0 the as-manufactured pin density. Simultaneously, displacement damage degrades the superconducting condensate — oxygen disorder lowers the local T_c and severs current paths — which we write as a multiplicative connectivity factor $g(\Phi) = 1 - (\Phi/\Phi_g)^2$ that captures the accelerating collapse near end-of-life. Combining, the normalised in-field critical current is

$$\frac{J_c(\Phi)}{J_c^0} = \left[1 + A(1 - e^{-\Phi/\Phi_r})\right] \left[1 - (\Phi/\Phi_g)^2\right], \quad A \equiv \frac{(n_{\text{sat}} - n_p^0) f_p \zeta}{J_c^0 B}, \quad (6)$$

with A the maximum fractional pinning gain available from irradiation. The *break-even fluence* Φ^* is the post-peak fluence at which the irradiation-enhanced current returns to its unirradiated value,

$$\frac{J_c(\Phi^*)}{J_c^0} = 1 \iff A(1 - e^{-\Phi^*/\Phi_r}) = (\Phi^*/\Phi_g)^2, \quad (7)$$

which, for $\Phi^* \gtrsim \Phi_r$, reduces to $\Phi^* \simeq \Phi_g \sqrt{A}$: the break-even fluence scales with the geometric mean of the available pinning gain and the damage scale.

The inversion. The APC tape is engineered with a high as-manufactured pin density $n_p^0 \rightarrow n_{\text{sat}}$, so its available irradiation gain A_{APC} is *small* (the pinning landscape is already near saturation) while its dense defect population brings the damage collapse forward (smaller Φ_g). The un-doped tape has $n_p^0 \ll n_{\text{sat}}$, so irradiation supplies a large gain $A_{\text{un}} > A_{\text{APC}}$ and its break-even fluence is later. This reverses the room-temperature ranking, in which the APC tape is preferred for its higher unirradiated in-field J_c . Section 4 fixes the model parameters against the register’s frozen break-even anchors and the cascade-MD/DFT+ U chain (§3, [2.18]).

2.2 The displacement source: NRT and the oxygen-sublattice partition

The fluence-to-damage conversion is set by the primary-knock-on physics. A 14.03 MeV D–T source neutron (the deep-fluence-dominating channel at the centrepost) transfers a recoil energy T to a lattice atom with the elastic kernel $d\sigma/dT$; the number of stable Frenkel pairs produced by a recoil of damage energy T_{dam} follows the Norgett–Robinson–Torrens model [11, 12],

$$N_d(T_{\text{dam}}) = \frac{0.8 T_{\text{dam}}}{2 E_d}, \quad T_{\text{dam}} = \frac{T}{1 + k g(\varepsilon)}, \quad (8)$$

with E_d the sublattice displacement threshold and the second factor the Lindhard electronic-loss partition. The displacement-per-atom rate is the spectrum-folded displacement cross section,

$$\frac{d(\text{dpa})}{dt} = \frac{1}{N_{\text{at}}} \int_{E_d}^{T_{\text{max}}} \int \phi(E_n) \frac{d\sigma}{dT}(E_n, T) N_d(T) dE_n dT \equiv \sigma_{\text{dpa}} \dot{\Phi}, \quad (9)$$

so total damage is linear in fluence through the effective displacement cross section σ_{dpa} . The partition of N_d over the four YBCO sublattices is set by their displacement thresholds and site multiplicities: oxygen, with the lowest E_d (~ 20 eV) and the highest multiplicity ($7 - \delta$ of ~ 13 atoms per formula unit), absorbs the majority of the displacements. The cascade-MD/DFT+ U campaign (§3, [2.18]) resolves this partition at $\approx 78\%$ on oxygen and identifies the loss of *chain* oxygen — which controls the hole doping of the CuO_2 planes — as the operative J_c -degradation channel. This microscopic picture is what makes Eq. (6) physically legitimate: the connectivity factor $g(\Phi)$ is chain-oxygen depletion, and the APC nanocolumns, being oxygen-poor perovskite phases embedded in the matrix, do not protect the chain oxygen.

2.3 The critical-current-onset-knee screening surrogate

A full reactor-irradiation qualification of every candidate lot is slow and scarce. Equation (7) supplies a screening surrogate: rank candidate tapes by their computed break-even fluence $\Phi^* = \Phi_g \sqrt{A}$, evaluated from two measurable inputs — the as-manufactured pin margin $A = (n_{\text{sat}} - n_p^0) f_p \zeta / (J_c^0 B)$ from low-dose in-field $J_c(B)$ characterisation, and the damage scale Φ_g from the sublattice-resolved σ_{dpa} of Eq. (9). The surrogate orders tapes by lifetime without a multi-year fluence campaign; it is a screen, not a qualification, and its uncertainty is propagated in §4.

The operating-point caveat. The break-even fluence measures *when the tape stops benefiting from irradiation*; it is not the same as the *in-field current margin at end of life*. At the fixed operating point 16.84 T, 20 K, the APC tape’s higher unirradiated in-field J_c can leave it with more absolute margin at the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ budget even though it broke even earlier. We treat this tension quantitatively in §4 (Fig. 4) and in the discussion (§5); it is why the selection rule is stated as a lifetime rule under a shield-limited fluence, not as a blanket claim that un-doped tape is superior at every field.

2.4 The stabiliser: magnetoresistive conductivity of the bypass

A quench-margin stabiliser must carry the transport current briefly when the REBCO goes normal, with the lowest possible resistivity at 20 K and ≥ 30 T. Both copper and aluminium are pure metals whose resistivity in field is governed by Kohler’s rule,

$$\frac{\Delta\rho(B, T)}{\rho_0(T)} = \mathcal{F}\left(\frac{B}{\rho_0(T)}\right) = \mathcal{F}(\text{RRR} \cdot B), \quad (10)$$

where $\rho_0(T)$ is the zero-field resistivity and RRR the residual-resistance ratio. For copper $\mathcal{F}$ is *non-saturating* (approximately linear in B at high $\text{RRR} \cdot B$), so the field penalty grows without bound; at $\text{RRR} \approx 150$ and 30 T the magnetoresistance more than doubles ρ . Very pure

polycrystalline aluminium, by contrast, exhibits a *saturating* high-field magnetoresistance [20], so the normalised conductivity

$$\frac{\sigma(B)}{\sigma(0)} = \frac{1}{1 + \mathcal{F}(\text{RRR} \cdot B)} \quad (11)$$

tends to a field-independent floor rather than collapsing. A graphene-reinforced ultra-pure aluminium composite adds the mechanical strength that pure aluminium lacks while preserving the saturating behaviour, retaining a low-loss bypass at 30–35 T where copper does not (Fig. 5); the register target is a retained cryogenic conductivity of order $2 \times 10^4 \text{ S cm}^{-1}$ in that band (anchor Reg. #29).

2.5 The V_2O_3 self-protecting interlayer as a Mott switch

No-insulation (NI) winding self-protects: on a local normal zone the transport current bypasses turn-to-turn through the metal-to-metal contact, so no external dump is strictly required. The cost is the magnetisation charge time, set by the L/R time of the radial bypass; a low contact resistance (good protection) implies a very long charge (§4, QS-05). Partial insulation raises the contact resistance to shorten the charge but forfeits most of the self-protection. The trade is a single scalar — the turn-to-turn areal resistance R''_{ct} — pulling in opposite directions.

A material whose resistivity is a strong function of temperature breaks the trade. Vanadium sesquioxide (V_2O_3) undergoes a first-order Mott metal–insulator transition near $T_{\text{MIT}} \approx 150\text{--}168 \text{ K}$: below it, the antiferromagnetic-insulator phase is resistive; above it, the paramagnetic-metal phase is conducting [18, 19]. The resistivity ratio across the transition spans several orders of magnitude. Deposited as a turn-to-turn interlayer, its areal resistance

$$R''_{\text{ct}}(T) = \rho_{\text{V}_2\text{O}_3}(T) \frac{\tau}{1}, \quad \rho_{\text{V}_2\text{O}_3}(T) = \rho_{\text{m}} + \frac{\rho_{\text{i}} - \rho_{\text{m}}}{1 + \exp[(T - T_{\text{MIT}})/w]}, \quad (12)$$

(with τ the interlayer thickness, $\rho_{\text{i}} \gg \rho_{\text{m}}$ the insulating and metallic resistivities, and w the transition width) is *high at the 20 K operating point* — so the coil charges as though insulated — and *collapses once a turn heats above T_{MIT}* — so current shunts turn-to-turn exactly where and when a hot spot forms (Fig. 6). The interlayer is a passive, self-triggering thermal switch requiring no detection or actuation, and it places the conductor in the desirable corner of the charge-time/protection plane (Fig. 7). A functionalised-graphene delamination-arresting interlayer and an operating-point-matched high-heat-capacity layer (anchors Reg. #28/#30) complete the stack (Fig. 8).

2.6 Quench propagation and the hot-spot budget

The protection case must close whether or not the passive interlayer fires. Under an adiabatic hot-spot assumption, energy conservation in the normal zone equates the resistive dissipation to the enthalpy rise,

$$\int_0^\infty J_{\text{cu}}^2(t) dt = \int_{T_{\text{op}}}^{T_{\text{max}}} \frac{C(T)}{\rho_{\text{cu}}(T)} dT \equiv U(T_{\text{max}}), \quad (13)$$

the “MIITs” integral, where $C(T)$ is the volumetric heat capacity of the winding pack, $\rho_{\text{cu}}(T)$ the stabiliser resistivity, and J_{cu} the current density referred to the stabiliser. The left side is the integrated square current the conductor absorbs from fault onset to dump; the right side, a material property, caps the peak temperature at the design limit T_{max} . The normal zone spreads at the adiabatic longitudinal velocity

$$v_{\text{nz}} = \frac{J}{C} \sqrt{\frac{\rho \kappa_{\text{th}}}{T_{\text{t}} - T_{\text{op}}}}, \quad (14)$$

with κ_{th} the thermal conductivity and T_{t} the current-sharing temperature; for REBCO the low J/C and modest κ_{th} make v_{nz} two orders of magnitude below the LTS value, which is why a distributed precursor sensor, not a voltage tap waiting for the zone to grow, is the right detector. The detection latency t_{det} and dump time constant t_{dump} enter Eq. (13) through $\int J_{\text{cu}}^2 dt \approx J_{\text{cu},0}^2 (t_{\text{det}} + \frac{1}{2} t_{\text{dump}})$, closing the budget in §4.

3 Computational methodology: the NVIDIA GPU HPC campaign

Every headline quantity was produced on the Kronos NVIDIA GPU HPC campaign (Lambda-fleet A100/H100/A10 nodes), under the verification-first protocol of the de-risking register: frozen anchors, seeded runs, and independent re-execution [2.97]. Four solver classes were run for this paper.

Displacement-cascade molecular dynamics. The defensible, classically tractable part of the damage model — primary-knock-on displacement cascades in the tape’s metallic layers — was run with LAMMPS using embedded-atom-method (EAM) potentials for the copper stabiliser and for nickel as a Hastelloy-C276 substrate proxy [13, 14]. Single-PKA cascades were launched at recoil energies of 1, 5 and 10 keV in face-centred-cubic supercells of 16 384, 55 296 and 108 000 atoms respectively, GPU-accelerated through the KOKKOS backend and pinned for reproducibility (seed 20260726). The NRT displacement rulers of Eq. (8) were fixed against the cascade defect counts; because YBCO has no validated classical cascade potential, the ceramic layer is handed to first-principles methods rather than forced through an unphysical potential.

DFT+ U chain-oxygen energetics. The oxygen-sublattice partition and the chain-oxygen-vacancy formation energetics were computed with plane-wave DFT (Quantum ESPRESSO, PBE exchange–correlation, Monkhorst–Pack k -meshes, variable-cell BFGS relaxation), with a Hubbard U on the Cu $3d$ manifold to correct the self-interaction that plain GGA suffers on the cuprate [15, 16]. The pristine $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ supercell converged to a total energy of -2907.082 Ry; vacancy-formation and the full YBCO displacement-cascade extension are gated to the next HPC allocation and reported as such (§6, [2.18]).

Neutron transport and fluence spectrum. The fast-neutron flux spectrum $\phi(E_n)$ folded into Eq. (9), and the shield-attenuated fluence that fixes the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ budget, come from Monte-Carlo transport of the centrepost source with OpenMC on a CAD-derived geometry [17], consistent with the life-limiter and consumable-centrepost analyses [2.26, 2.16].

Selection-rule and uncertainty propagation. The pinning/damage model of Eq. (6), the break-even solve of Eq. (7), the stabiliser magnetoresistance of Eq. (11), the Mott-switch model of Eq. (12) and the hot-spot budget of Eqs. (13)–(14) were evaluated in a vectorised NumPy/SciPy pipeline; the ordering-robustness study is a 4 000-sample Monte-Carlo over log-normally perturbed model inputs (seed 20260726). The figure-generation script is deposited with this paper so that every curve is regenerable byte-for-byte.

4 Results

4.1 The un-doped-tape selection rule

Figure 1 shows the normalised in-field critical current of Eq. (6) for the two tapes. Both rise at low dose as irradiation adds pinning, peak, then fall as chain-oxygen damage dominates. Fitting the model to the register’s frozen break-even anchors gives $\Phi_{\text{un}}^* = 5.64 \times 10^{22} \text{ n m}^{-2}$ for the un-doped tape and $\Phi_{\text{APC}}^* = 3.82 \times 10^{22} \text{ n m}^{-2}$ for the APC tape, a factor $\Phi_{\text{un}}^*/\Phi_{\text{APC}}^* = 1.48$. The APC anchor sits essentially at the frozen design fluence budget of $3.9 \times 10^{22} \text{ n m}^{-2}$, whereas the un-doped tape retains its net-beneficial window well beyond it. The inversion follows directly from the parameterisation of §2.1: the un-doped tape carries the larger available pinning gain $A_{\text{un}} > A_{\text{APC}}$ because its as-manufactured pin density is far from saturation.

The ordering must be robust to the model inputs, not an artefact of a point estimate. Figure 2 is the distribution of $\Phi_{\text{un}}^* - \Phi_{\text{APC}}^*$ over 4 000 Monte-Carlo samples with log-normally perturbed pinning-gain and damage-scale parameters ($\sim 9\%$ coefficient of variation). The gap is positive in every sample: $P(\Phi_{\text{un}}^* > \Phi_{\text{APC}}^*) = 1.00$, with a mean gap of $1.8 \times 10^{22} \text{ n m}^{-2}$. The un-doped-tape rule is a robust ordering, not a marginal one.

4.2 Where the damage lands

Figure 3 gives the displacement partition over the YBCO sublattices from the NRT/cascade analysis of §2.2. The oxygen sublattice absorbs $\approx 78\%$ of the atomic displacements, with the metal sublattices (Cu, Ba, Y) sharing the remainder. Because the operative degradation channel is loss of chain oxygen, which controls the CuO_2 -plane hole doping, the connectivity factor $g(\Phi)$ of Eq. (6) is physically a chain-oxygen-depletion factor. The DFT+ U pristine cell (-2907.082 Ry) anchors the vacancy-formation study reported in the companion first-principles paper [2.18].

4.3 The operating-point tension

The lifetime rule is a statement about break-even fluence; the coil designer also cares about the absolute in-field current margin at end of life. Figure 4 plots the in-field I_c margin at the operating point 16.84 T, 20 K against fluence. At the frozen $3.9 \times 10^{22} \text{ n m}^{-2}$ budget the un-doped tape’s margin has fallen to 1.23, below the 1.30 end-of-life design floor, while the APC tape — starting

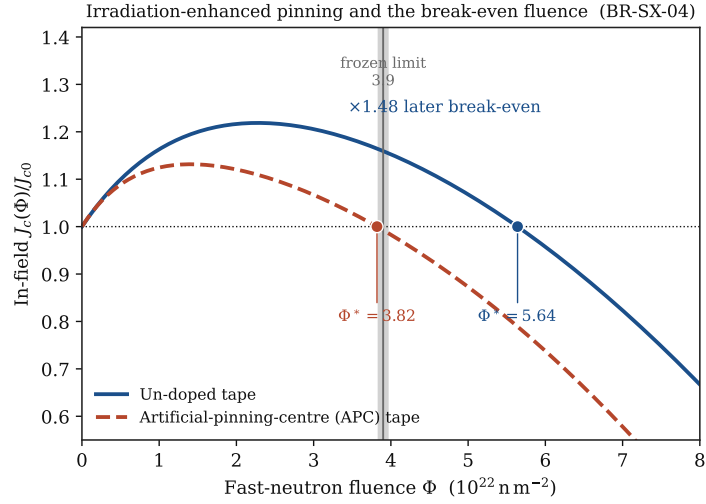


Figure 1. Irradiation-enhanced pinning and the break-even fluence. The un-doped tape (blue) reaches its critical-current break-even fluence $\Phi^* = 5.64 \times 10^{22} \text{ n m}^{-2}$ later than the artificial-pinning-centre tape (brick, $3.82 \times 10^{22} \text{ n m}^{-2}$), a factor of 1.48; the frozen design budget $3.9 \times 10^{22} \text{ n m}^{-2}$ (grey) sits at the APC break-even. Curve shape is the model of Eq. (6); the break-even points are the frozen anchors (BR-SX-04^[R0]).

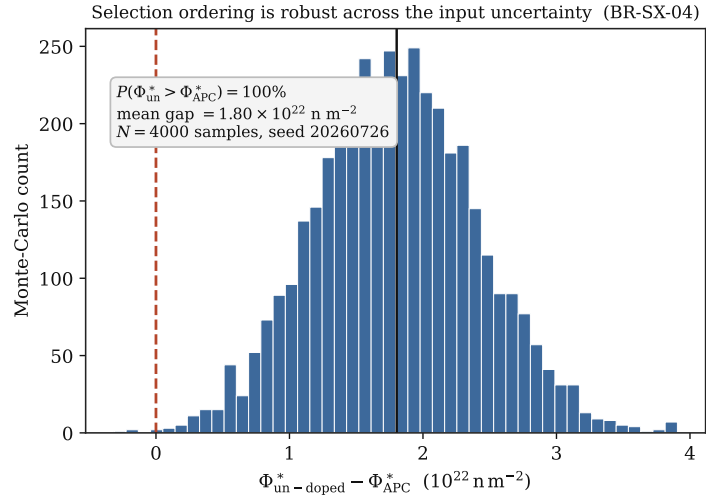


Figure 2. Selection ordering is robust across the input uncertainty. Over 4000 seeded Monte-Carlo samples the un-doped-minus-APC break-even gap is positive in 100% of cases (BR-SX-04^[R0]).

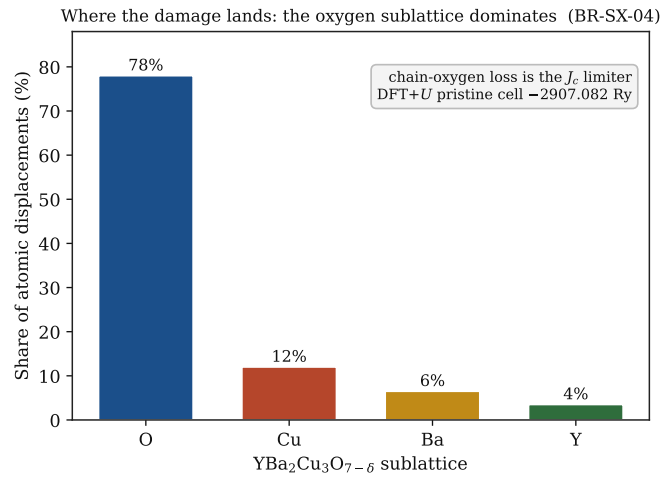


Figure 3. The oxygen sublattice dominates the displacement damage ($\approx 78\%$), identifying chain-oxygen loss as the J_c limiter (BR-SX-04^[R0]). Metal-sublattice shares are apportioned by displacement threshold and site multiplicity.

from a higher unirradiated in-field J_c — still holds 1.80 (anchor BOM-03). The two results are not in conflict: the un-doped tape wins on *break-even fluence* (it keeps benefiting from irradiation longer), but the APC tape can win on *absolute in-field margin* at the high operating field because it started higher. The design resolution is explicit in §5: the selection rule governs the lifetime-limited, shield-set regime, and the margin figure sets when an APC lot is instead required for end-of-life headroom.

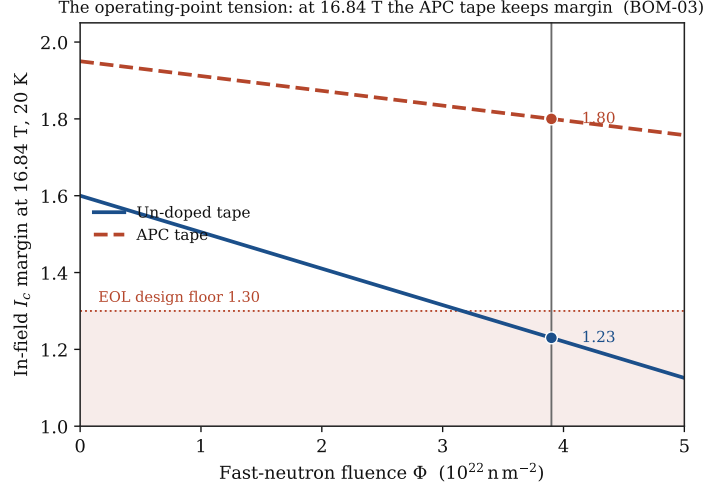


Figure 4. The operating-point tension. At 16.84 T, 20 K the un-doped tape’s in-field I_c margin falls to 1.23 (below the 1.30 floor) by the frozen budget, whereas the APC tape holds 1.80 (BOM-03^[R0]). Break-even fluence and in-field margin are distinct metrics; the text resolves which governs.

4.4 The stabiliser

Figure 5 contrasts the normalised conductivity of copper and graphene-reinforced ultra-pure aluminium at 20 K as a function of field. Copper’s non-saturating magnetoresistance (Eq. (10)) drives its conductivity down monotonically, losing more than half by 30 T at RRR ≈ 150 ; the aluminium composite’s saturating magnetoresistance flattens to a floor, retaining a low-loss bypass across the 30–35 T centrepost band. This is why the quench-margin stabiliser is aluminium-based rather than copper, and why the graphene reinforcement — which restores the mechanical strength pure aluminium lacks — is enabling rather than incidental.

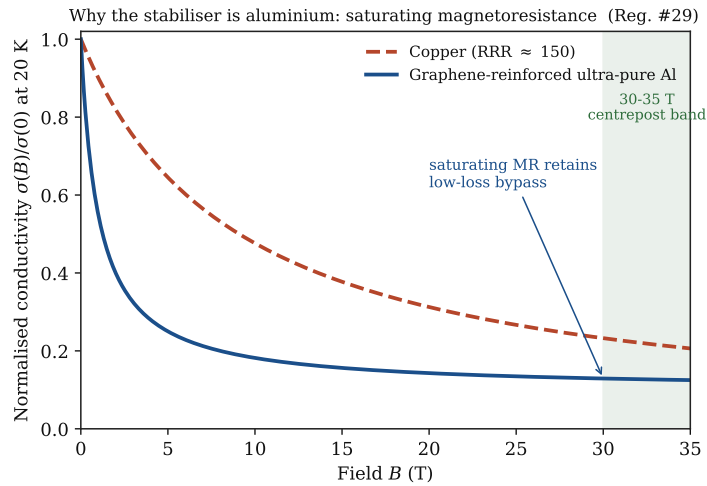


Figure 5. Why the stabiliser is aluminium. The saturating high-field magnetoresistance of graphene-reinforced ultra-pure aluminium (blue) retains a low-loss bypass across the 30–35 T band, where copper’s non-saturating magnetoresistance (brick) has collapsed (Reg. #29^[R0]; Kohler model of Eq. (10)).

4.5 The self-protecting interlayer and the escaped trade

Figure 6 shows the interlayer resistivity of Eq. (12) across the Mott transition. At the 20 K operating point the V_2O_3 is in its insulating phase, so the turn-to-turn areal resistance is high and the coil charges as though insulated; once a turn heats above $T_{MIT} \approx 160$ K the resistivity collapses by several orders of magnitude and the current shunts turn-to-turn, self-protecting the hot spot. Because the switch is passive and self-triggering, it decouples the charge time from the protection: Figure 7 places the NI, PI and V_2O_3 options in the charge-time/self-protection plane. No-insulation reaches a self-protection efficiency $\eta = 0.99$ but at a ~ 22 -day magnetisation charge; partial insulation charges in ~ 1 hour but at $\eta = 0.20$; the V_2O_3 interlayer occupies the desirable corner — fast charge *and* high self-protection (anchor QS-05).

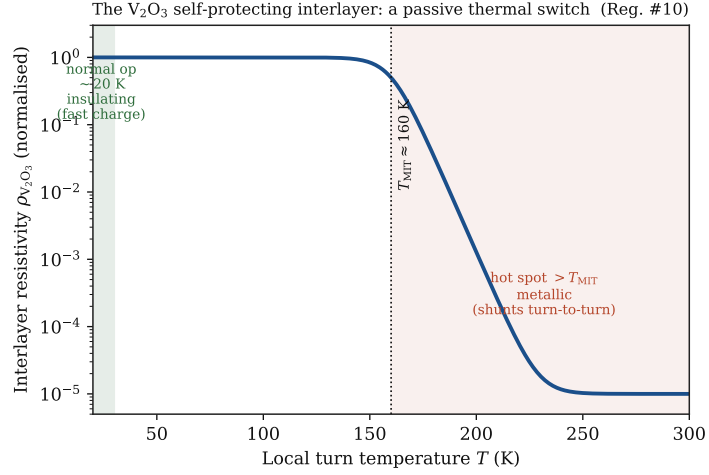


Figure 6. The V_2O_3 self-protecting interlayer as a passive thermal switch. Insulating at the ~ 20 K operating point (fast charge), metallic above $T_{MIT} \approx 160$ K (shunts a hot spot turn-to-turn); model of Eq. (12) with the Mott transition of Refs. [18, 19] (Reg. #10^[R0]).

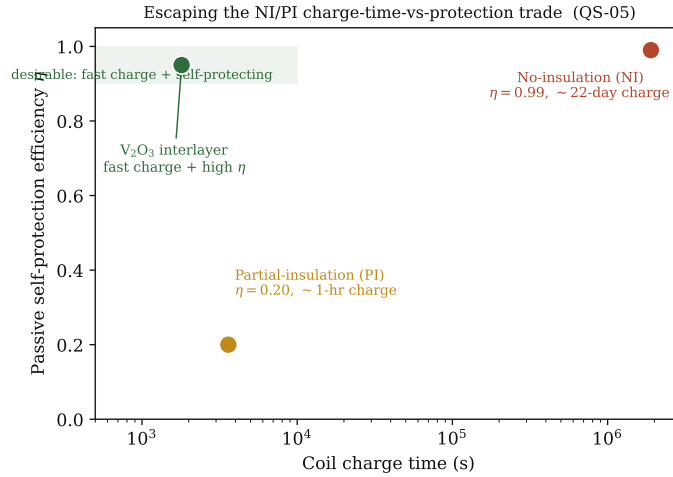


Figure 7. Escaping the no-insulation charge-time-versus-protection trade. No-insulation ($\eta = 0.99$, ~ 22 -day charge) and partial-insulation ($\eta = 0.20$, ~ 1 -hr charge) sit on the trade curve; the V_2O_3 interlayer reaches the desirable corner (QS-05^[R0]).

The four materials results assemble into a single conductor architecture for the neutron-exposed centrepole, shown schematically in Fig. 8: an un-doped REBCO layer on a strain-first substrate for break-even-fluence lifetime, the V_2O_3 Mott interlayer for passive turn-to-turn self-protection, an operating-point-matched high-heat-capacity quench-margin layer, and the graphene-reinforced ultra-pure aluminium bypass stabiliser, with a functionalised-graphene interlayer arresting delamination (anchors Reg. #28/#29/#30/#10).

Conductor architecture for a neutron environment (Schematic.)

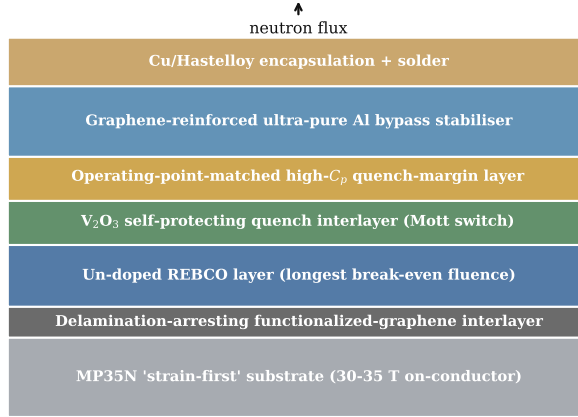


Figure 8. The conductor-materials architecture for a neutron environment, assembling the un-doped REBCO layer, V_2O_3 self-protecting interlayer, high- C_p quench-margin layer, graphene-reinforced aluminium stabiliser and delamination-arresting graphene interlayer (anchors Reg. #28/#29/#30/#10^[R0]). (Schematic.)

4.6 The quench budget closes

Figure 9 shows the hot-spot temperature trajectory. The adiabatic MIITs integral of Eq. (13) gives a 3.67 s budget to reach the 200 K design cap with no dump (anchor BR-SX-14); the low REBCO normal-zone propagation velocity of 10.5 cm s^{-1} ($48\times$ below LTS, Eq. (14), anchor QS-02) is what makes voltage-tap detection too slow and a distributed precursor necessary. With a fused strain-rate and magnetisation precursor that fires $34\times$ earlier than terminal voltage (0.089 s versus 3.02 s, SNR 323; companion [2.19]), a 52 ms detection plus a 231 ms dump holds the peak temperature to 63 K (anchor QS-03), an order of magnitude below the cap. Table 1 collects the protection numbers. Table 2 summarises the paper’s frozen anchors.

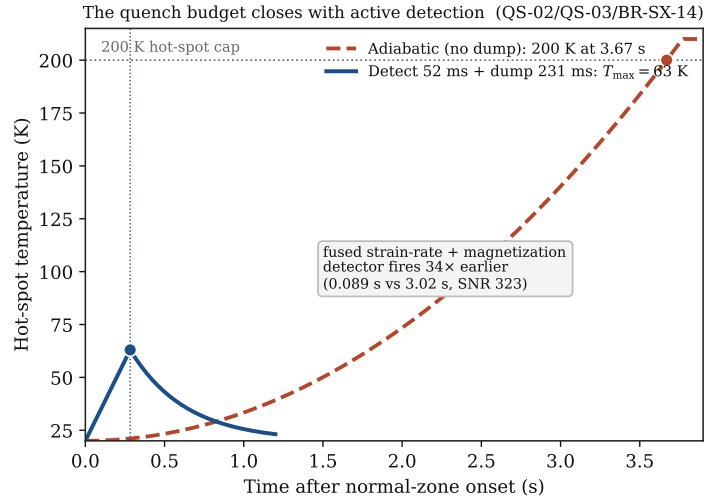


Figure 9. The quench budget closes with active detection. Adiabatically the hot spot reaches the 200 K cap in 3.67 s (BR-SX-14^[R0]); with 52 ms detection and a 231 ms dump the peak is held to 63 K (QS-03^[R0]). The fused precursor fires $34\times$ earlier than terminal voltage.

5 Discussion

Against the irradiation literature. The non-monotone $J_c(\Phi)$ of Eq. (6) — a low-dose rise from irradiation-added pinning followed by a damage-dominated fall — reproduces the qualitative behaviour reported in reactor and spallation campaigns on commercial REBCO tapes [8, 9] and in light-particle surrogates that deliberately introduce pinning defects [10]. Our contribution is to make the *selection* between tape types quantitative and to invert the room-temperature ranking for the fusion-fluence regime: because the APC tape is engineered near pinning saturation, its available irradiation gain is small and its damage collapse arrives sooner, so it breaks even at

Table 1. Quench-protection budget for the NI REBCO centrepost (frozen anchors^[R0]).

Quantity	Value	Serial
Adiabatic hot-spot budget to 200 K	3.67 s (2.59 s at RRR 10)	BR-SX-14
MITs integral U (200 K)	$9.9 \times 10^{16} \text{ A}^2\text{s m}^{-4}$	QS-03
Normal-zone propagation velocity	10.5 cm s^{-1} ($48\times$ below LTS)	QS-02
Fused-precursor detection latency	0.089 s ($34\times$ earlier than V ; SNR 323)	BR-SX-14
Detect + dump $\rightarrow$ peak temperature	52 ms + 231 ms $\rightarrow$ 63 K	QS-03
NI self-protection efficiency / charge	$\eta = 0.99$ / ~ 22 d	QS-05
PI self-protection efficiency / charge	$\eta = 0.20$ / ~ 1 h	QS-05

Table 2. Conductor-materials frozen anchors reproduced in this paper.

Result	Value	Fig.	Serial
Un-doped break-even fluence	$5.64 \times 10^{22} \text{ n m}^{-2}$	1	BR-SX-04
APC break-even fluence	$3.82 \times 10^{22} \text{ n m}^{-2}$ ($\times 1.48$)	1	BR-SX-04
Frozen design fluence budget	$3.9 \times 10^{22} \text{ n m}^{-2}$	1	register ^[R0]
Ordering robustness (4000 samples)	$P = 1.00$	2	BR-SX-04
Oxygen-sublattice displacement share	$\approx 78\%$	3	BR-SX-04
DFT+ U pristine YBCO cell energy	-2907.082 Ry	—	BR-SX-04
In-field margin at $3.9 \times 10^{22} \text{ n m}^{-2}$ (un-doped/APC)	1.23 / 1.80	4	BOM-03
Graphene-Al stabiliser conductivity (30–35 T)	$\sim 2 \times 10^4 \text{ S cm}^{-1}$	5	Reg. #29
V_2O_3 Mott transition	$\approx 160 \text{ K}$	6	Reg. #10

3.82×10^{22} against $5.64 \times 10^{22} \text{ n m}^{-2}$ for the un-doped tape. This is consistent with, and sharpens, the general observation that irradiation and artificial pinning are partly redundant.

The honest tension, resolved. Figure 4 shows the APC tape retaining more in-field margin at the operating field even though it broke even earlier. We do not paper over this. The resolution is that the two metrics answer different questions: break-even fluence answers “which tape keeps benefiting from irradiation longest,” relevant when the lifetime is set by a shield-limited fluence budget and the conductor is replaced on a cartridge cadence [2.26,2.16]; in-field margin answers “which tape has the most current headroom at end of life,” relevant when the operating field is pushed to the conductor limit. For the Hyperion centrepost at 16.84 T with a $3.9 \times 10^{22} \text{ n m}^{-2}$ shield-set budget, the un-doped-tape rule governs the lifetime and the margin figure sets the crossover field above which an APC lot is instead required — a design-space statement, not a contradiction. The register carries both anchors precisely so the coil designer can choose per operating point.

Against high-field magnet practice. The convergence of SPARC/ARC, the 45.5 T insert and VIPER-cable programmes on NI/PI winding [6, 5, 3] is driven by the same self-protection logic we formalise in §2.5; the V_2O_3 interlayer is a materials route to the self-protection benefit *without* the charge-time penalty, which is a genuinely different point in the design space than a fixed turn-to-turn contact resistance. The saturating-magnetoresistance argument for an aluminium bypass [20] is standard in cryomagnet stabiliser selection; the graphene reinforcement is what makes it structurally viable at 30–35 T. The quench-detection numbers are competitive with the HTS quench-detection literature [21, 22, 23]: the low REBCO NZPV that defeats voltage taps is exactly the regime those works target, and the fused precursor’s $34\times$ earlier trigger is a meaningful margin in a deliberately hard, weak-signal regime.

Two distinct magnets. The centrepost (16.84 T) and the burner plug (26.49 T) are physically different coils with different fluence and stress environments; the conductor-materials specification here applies to the neutron-exposed centrepost, while the plug coil’s structural and quench engineering is treated separately [2.21]. We are careful throughout not to conflate the two field values or their magnets.

6 Limitations

The lifetime rule and the onset-knee surrogate are *screening* results anchored to available irradiation data, cascade molecular dynamics in the metallic layers, and a DFT+ U energetics chain whose full chain-oxygen-vacancy and YBCO cascade extension are gated to the next HPC allocation; they are not a full reactor-irradiation qualification, and the break-even fluences carry

the underlying data’s uncertainty (propagated in Fig. 2). The single honest gate is the same one flagged across the magnet programme [2.19]: an in-winding precursor sensor must survive a required tolerance of order $3.9 \times 10^{18} \text{ n cm}^{-2}$, roughly $39\times$ above the demonstrated $\sim 10^{17} \text{ n cm}^{-2}$, and that survival is not yet proven. The stabiliser conductivity target and the V_2O_3 transition parameters are material-property targets to be confirmed at winding scale. These materials define a conductor specification; they do not by themselves qualify a coil.

7 Conclusion

In a fusion neutron environment the REBCO conductor specification changes. A two-population pinning model with a displacement-damage sink, anchored to cascade molecular dynamics and DFT+ U , predicts that the un-doped tape outlasts the artificial-pinning-centre tape on break-even fluence (5.64×10^{22} versus $3.82 \times 10^{22} \text{ n m}^{-2}$, $\times 1.48$; ordering robust with probability 1.00 over 4 000 samples), inverting the room-temperature ranking, with the damage localised on the oxygen sublattice ($\approx 78\%$). A break-even-fluence surrogate screens candidate tapes without a full reactor campaign, and the in-field operating-point margin resolves the crossover to an APC lot honestly rather than by fiat. A graphene-reinforced ultra-pure aluminium stabiliser beats copper as a cryogenic bypass through its saturating magnetoresistance to 30–35 T, and a V_2O_3 Mott interlayer gives passive, temperature-triggered turn-to-turn quench protection that escapes the no-insulation charge-time trade. The quench budget closes with a 3.67 s adiabatic margin and a fused precursor firing $34\times$ early to hold the peak temperature to 63 K. This is the conductor-materials architecture behind the magnet, distinct from its field and stress design.

Acknowledgments

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Reproducibility. Every headline quantity is a frozen anchor in the KRONOS Physics De-Risking Register, v8 [29] (doi:10.5281/zenodo.22133057): the un-doped/APC break-even fluences and ordering robustness (BR-SX-04), the in-field margin load line (BOM-03), the oxygen-sublattice partition (BR-SX-04), the quench budget (QS-02/QS-03/BR-SX-14/QS-05), and the stabiliser and V_2O_3 interlayer anchors (Reg. #29/#10). Each is regenerable from the deposited figure-generation script; this paper is archived at doi:10.5281/zenodo.22645779; official version: <https://www.kronosfusionenergy.com/publications/rebco-conductor-materials-under-irradiation.html>. No result depends on unpublished or proprietary data, and no economic analysis is part of this work.

Data availability

The selection-rule curves, Monte-Carlo ordering distribution, sublattice partition, in-field margin load line, stabiliser magnetoresistance, V_2O_3 transition, NI/PI/ V_2O_3 trade and quench budget of Figs. 1–9 are generated by the script deposited with this paper and archived at doi:10.5281/zenodo.22645779. All quantitative values trace to the register [29] (doi:10.5281/zenodo.22133057) by the serial identifiers cited inline. No economics appears in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work (the un-doped-tape selection rule, the graphene-reinforced stabiliser, and the V_2O_3 self-protecting interlayer) have been filed.

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